

# AlphaSkill: Verifier-Guided Self-Evolution of Process Skills for Formulaic Alpha Discovery

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## Abstract

Language-model agents can generate and repair formulaic alphas, yet their reusable capability is usually stored as factors, trajectories, free-form memories, or model parameters. None directly represents the conditional procedure that turns a market–pool–failure context into a productive sequence of research actions. We introduce ALPHASKILL, which learns such procedures as typed, versioned process skills while keeping the generator and evaluator fixed. Each attempt becomes a provenance-complete record; contrastive records propose skill edits; a contextual router composes accepted versions; and a chronological verifier promotes an edit only when paired meta-block evaluation improves a lower confidence bound without violating stability, redundancy, and cost guardrails. This design targets *evolutionary overfitting*: a chance backtest success being distilled into persistent future behavior. In a 20-seed planted/Null control, ALPHASKILL improves locked area-under-learning-curve (AULG) over Skill Reset by 0.0333 (95% CI [0.0251, 0.0416]) and reduces Null edit promotion from 100% to 3.1%. Across seven real-data episodes and ten paired seeds, China Carry improves AULG by 0.0082 ([0.0019, 0.0145]) and yields 15.0 additional non-redundant factors per 1,000 calls ([6.85, 23.15]). When U.S. evidence does not persist, the verifier accepts only 3 of 70 edits, all with positive locked gain. ALPHASKILL therefore makes self-evolution both cumulative and selective: it learns when evidence transfers and abstains when it does not.

## Introduction

Formulaic alpha discovery searches for interpretable programs that map market variables to predictive cross-sectional signals. Its central difficulty changes over time. Early search can exploit familiar momentum, reversal, volatility, and price–volume structures; later search must recognize which mechanisms are already represented, diagnose why new candidates fail, and reach complementary regions under a fixed evaluation budget. Symbolic and language-model systems explore effectively, but their reusable research behavior is usually implicit in populations, trajectories, memories, or model parameters.

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We instead learn the conditional research procedure that maps a market–pool–failure context to mining actions: diagnose a gap, select a hypothesis family, construct a typed expression, interpret rejection, and repair it. This procedure is more reusable than one factor and more operational than a retrieved episode. It has an applicability region, typed inputs and outputs, ordered actions, constraints, local tests, and a version history.

Finance makes this target unusually hazardous because backtests are noisy and repeatedly reused. A one-off false discovery produces one bad factor; a self-evolving system can additionally distill that accident into a persistent rule, route later search toward related accidents, and amplify selection bias across episodes. We call this *evolutionary overfitting*. ALPHASKILL addresses both sides of the problem: typed records make procedural evidence learnable, while disjoint verification decides which abstractions may persist.

Our contributions are:

- We formulate cumulative alpha discovery as learning conditional, versioned *process skills*, and identify evolutionary overfitting as a persistence-specific failure mode.
- We introduce a record-to-skill algorithm with contrastive compilation, contextual composition, seven structural lifecycle operators, and chronological lower-confidence-bound verification.
- We causally isolate skill accumulation through matched Carry/Reset replays and show positive China discovery-efficiency gains, calibrated U.S. abstention, and strong false-promotion suppression in planted/Null controls.

## Related Work

**LLM-driven formulaic alpha research.** Formulaic factors are executable and auditable (Kakushadze 2016). AlphaGen optimizes complementary collections (Yu et al. 2023); AlphaForge combines neural generation and dynamic combination (Shi et al. 2025); and trajectory-level shaping uses expert factors (Zhao et al. 2025). LLM systems add hypotheses and repair: AlphaAgent regularizes originality and alignment (Tang et al. 2025); RD-Agent(Q) co-develops factors and models (Li et al. 2025); Chain-of-Alpha separates generation from optimization (Cao et al. 2025); LLM-MCTS and AlphaPROBE organize trees or lineage (Shi, Duan, and Li 2025; Guo et al. 2026); and FactorEngine separates logic

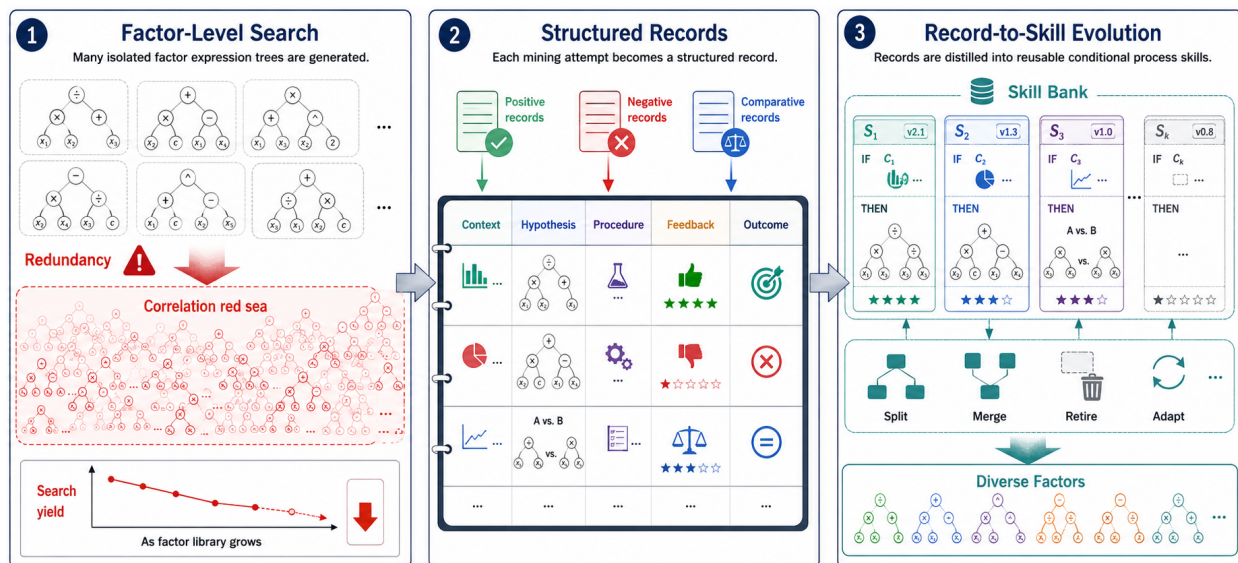


Figure 1: Attempt records are compiled into conditional process skills; only disjoint validation permits a version to affect later search.

Table 1: Representative agent-learning and alpha-mining systems. “Typed  $\pm$ ” means explicit positive, negative, and matched records; lifecycle denotes structural edits beyond appending memory.  $\Delta$  is partial support.

| Method                         | Persistent object   | Typed $\pm$ | Conditional proc. | Lifecycle | Disjoint gate | Carry/Reset |
|--------------------------------|---------------------|-------------|-------------------|-----------|---------------|-------------|
| AlphaAgent (Tang et al. 2025)  | Factor lineage      | —           | —                 | —         | —             | —           |
| FactorMiner (Wang et al. 2026) | Experience memory   | ✓           | $\Delta$          | —         | —             | —           |
| AlphaMemo (Yu et al. 2026)     | AST-edit memory     | ✓           | $\Delta$          | —         | —             | —           |
| Skill1 (Shi et al. 2026)       | SkillBank           | $\Delta$    | $\Delta$          | $\Delta$  | —             | —           |
| SkillOpt (Yang et al. 2026b)   | Text skill          | $\Delta$    | —                 | $\Delta$  | ✓             | —           |
| QuantaAlpha (Han et al. 2026)  | Research trajectory | $\Delta$    | —                 | $\Delta$  | —             | —           |
| ALPHASKILL                     | Process contract    | ✓           | ✓                 | ✓         | ✓             | ✓           |

from parameters (Lin et al. 2026b). These improve generators, trajectories, or expressions; ALPHASKILL learns the procedure controlling a frozen generator.

**Experience and procedural memory for agents.** Reflexion stores verbal feedback (Shinn et al. 2023), ExpeL extracts insights from success and failure (Zhao et al. 2024), Voyager accumulates executable code (Wang et al. 2023), and Agent Workflow Memory induces reusable routines (Wang et al. 2025). In alpha mining, FactorMiner records directions and failure constraints (Wang et al. 2026), while AlphaMemo models context-conditioned AST edits (Yu et al. 2026). These reduce repeated work, but an episode, insight, workflow, and skill differ in abstraction, and retrieval frequency does not prove future utility. ALPHASKILL preserves typed records, then tests compiled skills’ applicability and marginal value.

**Self-evolving skills and agents.** SkillRL distills hierarchical SkillBanks (Xia et al. 2026); Skill1 jointly learns selection, use, and distillation (Shi et al. 2026); and SkillOpt applies held-out bounded edits (Yang et al. 2026b). SkillOS curates repositories and SkillMaster optimizes counterfac-

tual probe utility (Ouyang et al. 2026; Yang et al. 2026a); MUSE-Autoskill manages the lifecycle (Lin et al. 2026a); and SkillFoundry packages scientific resources (Shen et al. 2026). Workflow/program search provides related external evolution (Zhang et al. 2025; Novikov et al. 2025); QuantaAlpha and AlphaAgentEvo bring it to alpha mining (Han et al. 2026; Tang et al. 2026). ALPHASKILL evolves a conditional, versioned process contract through create, refine, split, merge, compose, retire, and rollback, with promotion evidence separated from proposal records.

**Selection bias and validated financial discovery.** Reusing one history creates data snooping (White 2000); backtest winners can fail out of sample (Bailey et al. 2017); combining signals compounds bias (Novy-Marx 2015); and factor selection must balance false and missed discoveries (Harvey and Liu 2020). AlphaEval adds stability, robustness, logic, and diversity to predictive correlation (Ding et al. 2025). Self-evolution adds a persistence channel: noise can become procedure. We therefore use source records for proposals, disjoint meta blocks for promotion, locked blocks for effects, and pool-level metrics for redundancy.

## Problem Formulation

**Budgeted chronological discovery.** Discovery proceeds through chronological episodes  $e = 1, \dots, E$ . At episode  $e$ , the agent may access only data  $D_{\leq \tau_e}$  and receives a current factor pool  $\mathcal{L}_e$ , operator set  $\mathcal{O}$ , and budget  $B_e$ . A factor  $f \in \mathcal{F}$  is a typed executable expression over observable price–volume fields. The evaluator  $V$  is fixed before the experiment and returns a vector covering prediction, stability, robustness, logic, diversity, and cost. Pool utility is

$$J_e(\mathcal{L}) = w_p Q_{\text{pred}} + w_s Q_{\text{stab}} + w_r Q_{\text{rob}} + w_l Q_{\text{logic}} + w_d Q_{\text{div}}. \quad (1)$$

For an admitted set  $A_e$ , reward is the marginal improvement

$$r_e = J_e(\mathcal{L}_e \cup A_e) - J_e(\mathcal{L}_e) - \lambda C_e, \quad (2)$$

where  $C_e$  records model tokens, evaluator calls, money, and wall time.

**Records and skills.** Every attempted factor produces

$$R_i = \langle c_i, z_i, G_i, f_i, y_i, o_i, \ell_i, \kappa_i \rangle, \quad (3)$$

where  $c_i$  is the legally observable market–pool–budget context;  $z_i$  the research intent;  $G_i$  the invoked procedure trace;  $f_i$  the executable artifact;  $y_i$  evaluator feedback;  $o_i$  outcome;  $\ell_i$  lineage; and  $\kappa_i$  integrity metadata. A process skill is

$$s = \langle a_s, \pi_s, C_s, T_s, P_s \rangle, \quad (4)$$

where  $a_s$  specifies applicability,  $\pi_s$  an ordered procedure,  $C_s$  constraints,  $T_s$  local tests, and  $P_s$  versioned provenance and utility posteriors. A skill cannot modify  $V$ , the data boundary, or the final evaluation protocol.

**Valid self-evolution.** We call a SkillBank update valid self-evolution only if it is (i) *persistent* across episodes, (ii) *endogenous* to the agent’s execution evidence, (iii) *functional*, changing subsequent routing or execution, and (iv) *validated* on episodes not used to induce the update. Systems satisfying only the first three conditions are self-modifying, not necessarily self-improving.

## Method

One chronological episode implements five transitions: *execute* routed skills under a fixed budget; *record* every attempt and failure; *compile* recurring contrasts into candidate contracts; *verify* candidate versions against their parents on disjoint meta blocks; and *carry* only accepted versions into the next episode. This separation makes the evolving object, the evidence that changes it, and the evidence that judges the change independently inspectable.

## System State and Information Boundary

Mutable state is  $\Omega_e = (\mathcal{B}_e, \mathcal{S}_e, \Theta_e, \mathcal{L}_e)$ : typed records, versioned skills, router statistics, and the admitted pool. The base LLM, DSL, evaluator, preprocessing, split, and budget are frozen. Improvement can therefore arise only by changing available procedures, their selection, or execution—not the judge or data boundary.

Each fold assigns ordered roles  $D_e^{\text{src}} \prec D_e^{\text{meta}} \prec D_e^{\text{lock}}$ . Source outcomes propose edits; meta episodes gate them;

locked blocks are read only after decisions are frozen. A reported regression may schedule rollback for the next episode but cannot alter its score. Record cutoffs and lineage checks enforce this boundary.

At the start of an episode, a deterministic context encoder produces

$$c_e = [m_e, p_e, h_e, b_e], \quad (5)$$

where  $m_e$  contains lagged market descriptors,  $p_e$  pool coverage,  $h_e$  recent typed outcomes, and  $b_e$  remaining budgets. Routing uses continuous descriptors; hindsight regime labels are excluded.

## Typed Record Construction

After execution,  $c_i$  stores legal context;  $z_i$  the hypothesis and target role;  $G_i$  the ordered skill/action/tool trace;  $f_i$  the canonical AST;  $y_i$  pre-admission feedback;  $o_i$  the accepted/rejected/failed outcome; and  $\ell_i$  lineage. Integrity metadata  $\kappa_i$  binds fold, cutoff, pool snapshot, seed, model/prompt and evaluator hashes, configuration, and cost.

Failures map to a fixed taxonomy: syntax/type, illegal access, invalid numerical output, weak prediction, instability, perturbation sensitivity, redundancy, semantic inconsistency, or budget violation. Negative records preserve the failed procedure and checks, because similar RankIC may conceal different failure mechanisms.

**Comparative records.** A ComparativeRecord requires matched context, target, budget, model, seed block, and evaluator, with one declared procedural difference. Child/parent attempts store  $\Delta G_{i,i'}$ ,  $\Delta y_{i,i'}$ , and  $\Delta J_{i,i'}$ . Pairs with uncontrolled differences remain ordinary records but receive no causal skill credit.

Construction is deterministic and append-only. Annotations may add links or status but cannot overwrite context, trace, artifact, or outcome; immutable-field hashes expose silent changes.

## Contrastive Record-to-Skill Compiler

Sufficient new support, repeated failures, or effect heterogeneity triggers compilation. Records are grouped by intent, compatible operator family, and context neighborhood. The compiler mines action subsequences enriched among positives, reversed among negatives, and supported by comparative pairs. A contrast  $q$  must occur in at least  $n_{\min}$  independent context/time blocks and satisfy

$$\hat{g}(q, A) = \frac{1}{|\mathcal{P}_{q,A}|} \sum_{(i,i') \in \mathcal{P}_{q,A}} \Delta J_{i,i'} > 0, \quad (6)$$

for a candidate applicability region  $A$ . The proposal stage uses this statistic only to prioritize compilations; it is not evidence for promotion.

A constrained LLM converts the contrast into Equation 4: source-specific fields, windows, and AST nodes become typed arguments; preconditions become predicates; and actions become executable steps. The contract includes stable identity/version,  $a_s(c)$ , typed I/O,  $\pi_s$ , limits  $C_s$ , tests  $T_s$ , and provenance  $P_s$ .

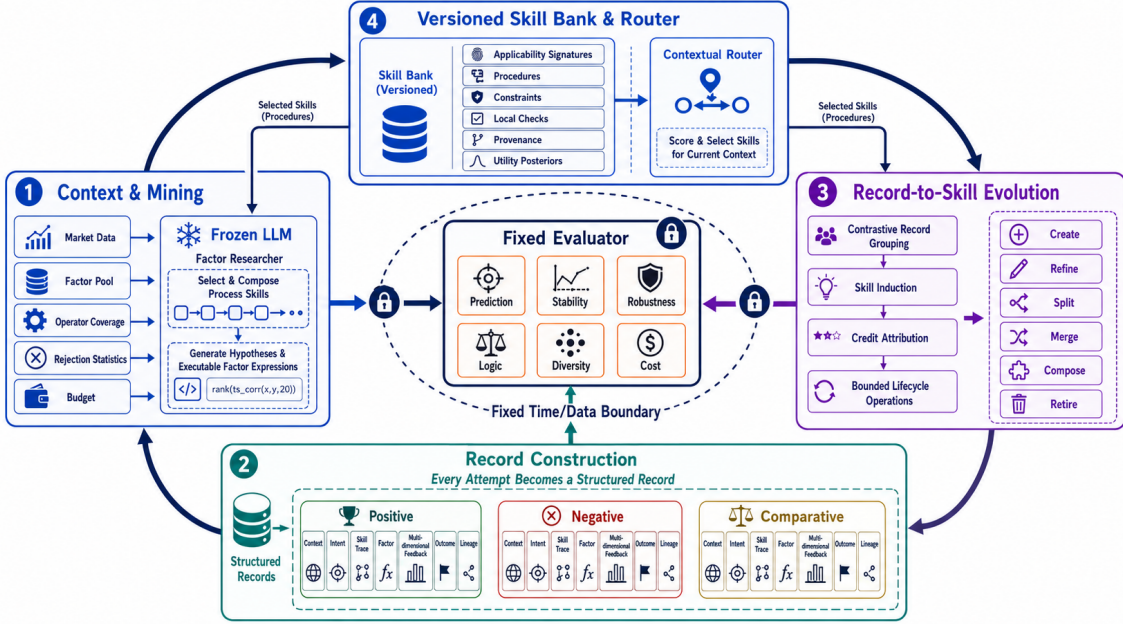


Figure 2: ALPHASKILL. The base LLM and evaluator remain frozen. Attempts become typed records; contrastive record groups propose skill edits; matched validation gates version promotion; the router reuses accepted skills in later chronological episodes.

To prevent factor memorization, compilation removes concrete expressions and returns, a de-memorization test rebinds held-out fields/operators, and contracts may store principles but not privileged source factors. Output must parse against the Skill schema and use registered APIs.

**Learning applicability by splitting.** Observable-context predicates expose heterogeneous effects. If  $\hat{g}(q, A_1) > 0$  but  $\hat{g}(q, A_2) < 0$ , the shallowest sufficiently supported stable-sign split produces context-specific children. Applicability is thus learned, not appended as a retrieval tag.

### Contextual Skill Routing and Composition

Eligible skills require  $a_s(c_e) = 1$ , available inputs, compatible constraints, and affordable cost. A Bayesian linear utility model over  $\phi(c_e)$  supplies a Thompson draw  $\tilde{\theta}_s$  and

$$u_e(s) = \phi(c_e)^\top \tilde{\theta}_s + \gamma N(s, \mathcal{L}_e) - \eta C(s), \quad (7)$$

where  $N$  is novelty relative to pool coverage and  $C$  expected cost. Under-supported skills use a conservative prior.

At most  $K_s$  skills form a typed Diagnose  $\rightarrow$  Hypothesize  $\rightarrow$  Construct  $\rightarrow$  Repair/Validate  $\rightarrow$  Admit DAG. Compatible DAG selection approximates

$$\max_G \sum_{s \in G} u_e(s) - \rho \sum_{s \neq s'} \text{Red}(s, s') \quad \text{s.t.} \quad C(G) \leq b_e, \quad (8)$$

where Red penalizes procedural overlap. A greedy solver suffices for small  $K_s$ . The frozen LLM receives legal context, selected contracts, pool descriptors, and the DSL—never source returns or locked outcomes.

### Credit Assignment

The routed DAG receives online advantage  $A_e = r_e - \hat{b}(c_e)$  from paired same-context/budget runs. It updates routing but cannot rewrite a skill: reward may be noisy and shared across nodes. Candidate and parent versions are instead re-played with matched context, order, seeds, and evaluator. Meta-block gain is

$$d_{s,j} = J_j(\mathcal{L}_j^s) - J_j(\mathcal{L}_j^{\text{parent}(s)}). \quad (9)$$

Thus cheap online credit routes, while paired intervention promotes versions, avoiding both Shapley cost and equal-credit assumptions.

### Verifier-Gated Skill Lifecycle

Seven bounded operations cover CREATE, REFINE, conditional SPLIT, compatible MERGE, sub-DAG COMPOSE, lineage-preserving RETIRE, and atomic ROLLBACK. None may modify the LLM, evaluator, DSL, data roles, or reporting rule.

Static checks enforce schema, types, provenance, hashes, and budgets; behavioral checks run unit, adversarial-binding, and de-memorization tests. The statistical gate uses  $H$  non-overlapping meta blocks, avoiding factor-level pseudo-replication. With mean  $\bar{d}_s$  and deviation  $\hat{\sigma}_s$ , its one-sided bound is

$$\text{LCB}_{1-\alpha}(s) = \bar{d}_s - t_{1-\alpha, H-1} \frac{\hat{\sigma}_s}{\sqrt{H}}. \quad (10)$$

The verifier promotes a candidate only if

$$\text{LCB}_{1-\alpha}(s) > \delta \quad \wedge \quad \Delta g_{s,q} \geq -\epsilon_q \quad \forall q, \quad (11)$$

where  $\delta$  is a minimum effect and  $g_q$  guard instability, robustness, correlation, invalid outputs, and cost. Rejected versions remain as evidence.

Accepted edits create immutable nodes with parents, support, validation blocks, effects, and guardrail deltas. Child exposure is initially capped. Locked outcomes only measure precision and schedule later rollback; retirement bounds the active library without erasing history.

**Proposition 1 (persistence amplification).** Consider  $K$  candidate edits with zero true gain whose reused-backtest estimates are independent  $\mathcal{N}(0, \sigma^2)$ . Accepting the largest estimate above  $\tau$  gives false-promotion probability

$$\Pr\left(\max_{k \leq K} \hat{d}_k > \tau\right) = 1 - \Phi(\tau/\sigma)^K, \quad (12)$$

which approaches one as edits accumulate. More generally, for  $\sigma$ -sub-Gaussian noise, a union bound requires  $\tau \geq \sigma\sqrt{2 \log(K/\alpha)}$  to keep this probability below  $\alpha$  (proof in the supplement). A self-revising system therefore faces a multiplicity cost that grows with its number of attempted edits, even when its generator is unchanged. Chronologically disjoint evidence, block-level uncertainty, and the lower-confidence-bound gate prevent selection noise from becoming a durable search prior.

## Complexity and What Constitutes Evolution

Appending a record is  $O(1)$  beyond its trace; routing is  $O(|\mathcal{S}_e|d + K_s^2)$ . Periodic compilation uses one constrained LLM call per supported contrast, and verification adds  $2H$  matched executions. Retire/merge bound active state.

RecordBank alone is memory; a changed file alone is self-modification. Self-evolution additionally requires endogenous persistent change, changed later behavior, and disjoint validation. Resetting skills while retaining records isolates evolution; removing the gate tests selection noise.

## Experiments

### Experimental Design

We ask three questions. **RQ1:** Does carrying verified process skills improve locked factor-pool utility and non-redundant discovery yield? **RQ2:** Does the verifier distinguish reusable edits from persistent selection noise? **RQ3:** Do improved discovery dynamics translate into economic value under one frozen portfolio rule?

**Markets and chronology.** We use two public 2020–2025 panels: 50 China A-shares fixed by market value on 2020-01-02 (72,595 symbol-days) and a predeclared 50-stock U.S. large-cap universe (75,400 symbol-days) (Ellendan 2026; Papers With Backtest 2026). Seven six-month episodes span 2022H2–2025H2. Each episode uses 24 months of source data, six months of meta data, and a disjoint six-month locked block; the final five signal dates of every role are purged. Five-day returns align evaluation with five-day rebalancing.

**Causal replay and baselines.** The study freezes proposal generation to 122 typed expressions, making differences attributable to process state, routing, or verification. *Random*

Table 2: Frozen real-data replay protocol.

| Dimension  | Setting                                           |
|------------|---------------------------------------------------|
| Chronology | 7 episodes; 24/6/6-month source/meta/locked       |
| Generator  | 122 typed expressions; 16 economic families       |
| Budget     | 20 calls/episode; 10 paired seeds                 |
| Target     | 5-day cross-sectional forward return              |
| Admission  | RankIC, stability, validity, and corr. $\leq .85$ |
| Portfolio  | Frozen nonnegative Ridge; rebalance every 5 days  |
| Costs      | 10 bps China; 5 bps U.S., one way                 |

samples uniformly; *Episodic UCB* adapts then resets; *Global Memory* carries family outcomes; *Skill Reset* retains records but restores seed skills; *w/o Gate* carries every edit; and *ALPHA SKILL* carries only verified edits. All share candidate order, noise, evaluator, admission, combination, dates, and calls. Table 1 locates full LLM-agent systems; the replay is a matched mechanism comparison.

**Carry/Reset identification.** Carry persists accepted versions; Reset restores the same seed SkillBank while retaining the same RecordBank. Common random numbers, contexts, and budgets define

$$\text{AULG} = \frac{1}{E} \sum_{e=1}^E [J_e^{\text{locked}}(\mathcal{S}_e^{\text{Carry}}) - J_e^{\text{locked}}(\mathcal{S}_1^{\text{Reset}})] . \quad (13)$$

Thus neither a stronger seed prompt nor additional search can masquerade as evolution.

**Metrics and statistics.** Primary endpoints are locked AULG and admitted non-redundant factors per 1,000 calls (MY). Secondary measures include RankIC, ICIR, pool correlation, annualized excess return (ARR), information ratio (IR), maximum drawdown (MDD), update precision, and future regression. Ten paired seeds give 95%  $t$  intervals; daily differences use 4,000 moving-block bootstrap replicates with 20-day blocks. Update precision is the fraction of activated edits with positive never-selected locked gain.

**Audit trail.** All 16,800 real DiscoveryRecords and 280 EvolutionRecords bind the factor AST, fold and cutoffs, pool snapshot, seed, configuration and evaluator hashes, decision, skill version, and cost. A separate controlled study contributes 15,360 DiscoveryRecords and 640 EvolutionRecords with explicit world labels; aggregation rules prevent controlled records from entering real-market tables.

### RQ1: Verified Skills Improve Discovery Efficiency

Table 3 establishes that the replay mines genuine China signals: ALPHA SKILL obtains locked RankIC 0.0473 ([0.0324, 0.0622]) and ICIR 0.184. More importantly, the paired identification in Table 4 shows what evolves. Carry raises AULG by 0.0082 ([0.0019, 0.0145],  $p = .016$ ) and MY by 15.0 factors per 1,000 calls ([6.85, 23.15],  $p = .002$ ). It also exceeds Global Memory AULG by 0.0070 ([0.0013, 0.0127]). The final RankIC difference from Reset is smaller and includes zero (+0.0013, [−0.0014, 0.0040]):



Table 3: Locked factor-pool quality under 20 candidate calls per episode. Values average seven chronological episodes and ten paired search seeds. Corr is mean absolute within-pool correlation; MY is admitted non-redundant factors per 1,000 calls; AULG is area under episode-level locked utility.

| Method              | China fixed universe |      |      |       |       | U.S. fixed universe |      |      |       |       |
|---------------------|----------------------|------|------|-------|-------|---------------------|------|------|-------|-------|
|                     | RankIC               | ICIR | Corr | MY    | AULG  | RankIC              | ICIR | Corr | MY    | AULG  |
| Random Search       | 0.0452               | .184 | .149 | 136.4 | .1105 | .0021               | .016 | .161 | 123.6 | .0231 |
| Episodic UCB        | 0.0457               | .184 | .173 | 142.9 | .1107 | -.0001              | .007 | .184 | 122.1 | .0230 |
| Global Memory       | 0.0473               | .188 | .195 | 148.6 | .1119 | .0000               | .006 | .182 | 121.4 | .0231 |
| Skill Reset         | 0.0460               | .185 | .179 | 140.0 | .1107 | -.0001              | .007 | .183 | 121.4 | .0230 |
| ALPHASKILL w/o Gate | 0.0498               | .186 | .317 | 188.6 | .1303 | .0000               | .004 | .201 | 123.6 | .0198 |
| ALPHASKILL          | 0.0473               | .184 | .194 | 155.0 | .1189 | -.0003              | .006 | .184 | 121.4 | .0226 |

Locked real-market replay (mean over 10 paired search seeds)

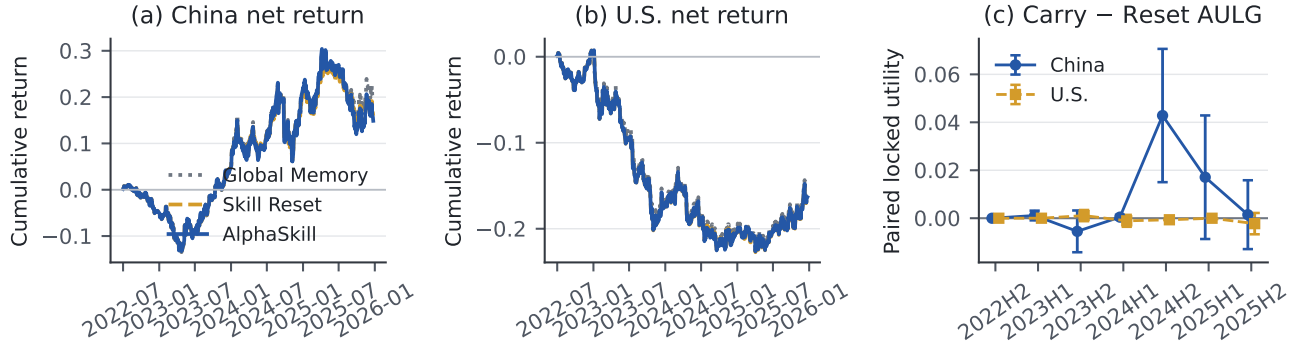


Figure 3: Locked fixed-universe replay. Panels (a–b) concatenate non-overlapping locked blocks and average daily net returns over ten search seeds. Panel (c) reports the paired Carry–Reset AULG difference (mean and 95% seed-level interval) by episode.

Table 4: Primary paired Carry–Reset effects. Intervals are seed-level 95% CIs; MY is per 1,000 calls.

| Market | Metric | $\Delta$ | 95% CI          |
|--------|--------|----------|-----------------|
| China  | RankIC | .0013    | [−.0014, .0040] |
| China  | AULG   | .0082    | [.0019, .0145]  |
| China  | MY     | 15.0     | [6.85, 23.15]   |
| U.S.   | RankIC | −.0002   | [−.0007, .0002] |
| U.S.   | AULG   | −.0004   | [−.0013, .0005] |
| U.S.   | MY     | 0.0      | [0.0, 0.0]      |

verified skills improve the *arrival rate* of complementary factors rather than manufacturing a single dominant expression.

This distinction is visible in the ungated condition. It reaches higher raw China MY, but mean pool correlation rises from 0.194 to 0.317 and its activated-edit regression rate nearly doubles. In the U.S., where locked RankIC is  $−0.0003$  ( $[−0.0068, 0.0062]$ ), Carry does not improve AULG. The method does not force accumulation: unpersuasive evidence leaves the SkillBank nearly unchanged.

**Accumulation across episodes.** Panel (c) of Figure 3 shows that China Carry–Reset gains emerge after verified versions accumulate, peaking in 2024H1–H2 rather than appearing at initialization. Across 10/20/40 calls, China AULG

Table 5: Locked rank-proportional long–short portfolios. ARR and MDD are percentages. Weights are fit on source+meta only, rebalanced every five trading days, with one-way costs of 10/5 bps in China/U.S.

| Method | China |      |       | Method | U.S. |       |       |
|--------|-------|------|-------|--------|------|-------|-------|
|        | ARR   | IR   | MDD   |        | ARR  | IR    | MDD   |
| Random | 4.4   | 0.31 | −9.9  | Random | −4.3 | −0.42 | −9.7  |
| UCB    | 6.3   | 0.46 | −9.5  | UCB    | −5.1 | −0.46 | −10.1 |
| Memory | 6.2   | 0.45 | −9.6  | Memory | −4.9 | −0.47 | −10.0 |
| Reset  | 5.3   | 0.39 | −9.5  | Reset  | −5.1 | −0.46 | −10.1 |
| w/o G  | 5.7   | 0.38 | −10.0 | w/o G  | −3.9 | −0.34 | −9.5  |
| AS     | 5.1   | 0.35 | −9.8  | AS     | −5.1 | −0.46 | −10.1 |

deltas remain positive (0.0184/0.0082/0.0053); the ten-seed main budget resolves the effect most precisely. U.S. deltas remain centered near zero at all three budgets. This temporal shape rules out an initially stronger skill prompt as the source of the China gain.

## RQ2: Verification Converts Evolution into Selective Learning

Table 7 reports behavior on exactly the objects that persist. In China, the verifier promotes 12 of 70 proposals; 11 im-

Table 6: Controlled process-learning mechanisms over 20 paired seeds.  $\Delta$  compares gated Carry with Skill Reset.

| World   | Measure                | $\Delta$ /Rate | 95% CI         |
|---------|------------------------|----------------|----------------|
| Planted | AULG $\Delta$          | .0333          | [.0251, .0416] |
| Planted | Final utility $\Delta$ | .0224          | [.0145, .0302] |
| Planted | MY $\Delta$            | 70.3           | [58.6, 82.0]   |
| Planted | Update precision       | 100.0%         | —              |
| Null    | Promotion, gated       | 3.1%           | —              |
| Null    | Promotion, no gate     | 100.0%         | —              |

Table 7: Skill-update behavior. P is proposal promotion; UP/Reg. are positive/negative locked gain among activated edits.

| Market | Variant | P     | UP    | Reg. |
|--------|---------|-------|-------|------|
| CN     | No gate | 100.0 | 84.3  | 15.7 |
| CN     | Gated   | 17.1  | 91.7  | 8.3  |
| US     | No gate | 100.0 | 47.1  | 52.9 |
| US     | Gated   | 4.3   | 100.0 | 0.0  |
| Null   | No gate | 100.0 | 50.0  | 50.0 |
| Null   | Gated   | 3.1   | 40.0  | 60.0 |

prove never-selected locked utility, giving 91.7% precision and 8.3% regression. The ungated variant promotes all 70 and activates 11 regressions. In the U.S., the gate promotes only 3 of 70 edits, all positive, whereas ungated evolution activates 37 regressions. Low promotion is therefore calibrated abstention, not a failure to evolve.

The controlled experiment in Table 6 separates successful learning from market idiosyncrasy. Across 20 paired seeds and recurring planted regimes, Carry improves locked AULG by 0.0333 ([0.0251, 0.0416]), final utility by 0.0224 ([0.0145, 0.0302]), and MY by 70.3 ([58.6, 82.0]), with 100% locked update precision. Under label-independent Null feedback, promoting every proposal yields 100% promotion; the disjoint gate reduces it to 3.1%. Together these results validate both halves of the algorithm: process state accumulates reusable structure when it exists, and the verifier blocks its persistence channel when it does not.

### RQ3: Economic Value under a Frozen Portfolio Rule

All methods use the same nonnegative Ridge combination fitted on source+meta and frozen on locked dates. Table 5 and Figure 3 show that ALPHASKILL’s China pool retains positive economic value: 5.1% net ARR, IR 0.35, and  $-9.8\%$  MDD at 10 bps one-way cost. Repricing the same positions gives 7.8% ARR before costs, 5.0% at 10 bps ([0.2, 9.9]), and 2.3% at 20 bps ([ $-2.6$ , 7.1]). Because UCB and Global Memory have higher point IR, the economic experiment supports viable factor-pool value, while the causal advantage of ALPHASKILL remains discovery efficiency.

U.S. portfolios are negative even before realistic costs; ALPHASKILL records  $-5.1\%$  ARR and IR  $-0.46$  at 5 bps. This outcome is consistent with the near-zero RankIC and near-total rejection of skill edits. The evaluator, verifier, and

portfolio therefore agree on where the learned process is useful.

## Discussion

**What the system learns.** The strongest China effect appears in AULG and marginal yield rather than terminal RankIC. This is the signature expected from process learning: a skill reallocates a fixed budget toward productive factor families earlier, but it does not change the best expression already present in the common bank. Family-level traces support this mechanism. The carried router attempts all 16 families yet concentrates successful admissions in low-volatility (96/135), range (26/127), and price-volume (20/106) structures, while repeatedly unproductive intraday and residual regions receive less attention.

**Why selective non-evolution matters.** Most self-evolving-agent evaluations reward the amount of change. ALPHASKILL instead treats rejection as a first-class outcome. The U.S. and Null results show why: persistence is valuable only when an edit’s advantage survives chronological separation. An agent that refuses 67 of 70 U.S. edits, while retaining all source evidence for later contexts, is better calibrated than one that continuously rewrites itself to fit weak feedback.

**Scope of the evidence.** Predeclared fixed universes and a frozen proposal bank isolate process-state evolution from constituent churn, generator updates, and portfolio-model differences. The resulting claim is precise: under a common expression space and budget, verified skills improve China discovery dynamics and suppress unsupported persistence. Replay returns are evidence of pool utility rather than deployable live alpha; impact, borrow, and capacity are outside the shared evaluator.

## Conclusion

ALPHASKILL moves self-evolving alpha discovery from accumulating artifacts to learning testable research procedures. Typed records preserve evidence, contrastive compilation creates conditional contracts, contextual routing changes later search, and chronological verification decides which versions persist. Matched replays show that verified skills improve China learning dynamics and non-redundant yield; planted/Null controls show that the same gate sharply suppresses evolutionary overfitting; U.S. results show calibrated abstention when evidence does not persist. The central lesson is that self-evolution is defined as much by disciplined refusal as by cumulative change.

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